Introduction

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Poorly designed roads lead to traffic accidents, which often lead to fatalities. However, not all roads and intersections are equally dangerous: some have no fatalities in the average year, while others can have ten or more. Given the number of factors potentially influencing road accidents, this is an area ripe for investigation using machine learning methods. Knowing more about design choices that influence intersection safety would be particularly useful for urban planning. In 2022, about 28.3% of traffic fatalities involved an intersection, and around 35% of those happened at signalized intersections (Federal Highway Administration, 2024). By identifying factors that contribute to the number of traffic accidents, researchers can help urban planners design safer intersections.

There have already been many efforts to use machine learning methods to predict accident occurrence and severity. Kassu & Hassan (2019) performed a PCA of fourteen factors, and boiled them down to seven dimensions incorporating factors such as number of lanes, weather, and driver demographics. Chen et al. (2012) found seven conditions that influenced accident risk: age and gender of driver, whether the accident was in a speed zone, traffic control type, time of day, crash type, and seat belt usage. Li (2025) did a similar thing with crash severity in the UK.

There are many ways this work could be expanded. For instance, in their investigation of signalized intersections in Korea, Yang et al. (2024) stressed the need for future investigations of unsignalized intersections. On the other side, Abdel-Aty & Haleem (2011) focused on developing machine learning models to identify connections between road features and accident occurrences at unsignalized intersections. Looking at commonalities and differences between signalized and unsignalized intersections would be beneficial as well.

Notably, most of the above cited studies were interested in *all* possible factors going into a crash, including both features of the road, temporary environmental conditions such as weather, and driver-related factors. This is of limited use when trying to learn what kind of intersection designs are safest. More work focusing purely on road design could be of use to urban planners.

Complicating this, other studies suggest that the metadata usually collected for these models may not capture key accident predictors. Lin (2022) analyzed accidents occurring in Taiwan at 3 way intersections. They found two categories of factors that strongly affected risk of accidents: specific features of the road, such as shoulder and lane width, and features in the surrounding environment, such as gas stations and supermarkets. Both of these highly relevant categories include information not usually included in intersection metadata databases. Certainly going forward, data collection about intersections can make sure to track this relevant information. However it is still important to find ways to work with the data we have now.

What's more, there may be important components of intersection safety no one has identified yet. No matter how broad we make our collection of metadata, we will never entirely eliminate the risk that we are neglecting to track some essential predictor.

One potential way to combat this risk and expand the predictive power of intersection safety models is to incorporate street-level images of intersections into our models. These images will naturally include many factors not captured by relevant metadata, such as the width of the roads, and features of the environment such as nearby gas stations. This could potentially allow a model to pick up on relevant features that handpicked metadata might miss, and lead to more accurate predictions.

A major limitation of using intersection image data is the sacrificed interpretability. With a metadata model, we can easily analyze which of the input factors are contributing to the model's output. With image data, this becomes much more difficult. On the other hand, analyzing commonalities between images flagged as high-risk may be a step towards identifying safety-relevant features of intersections that are not yet on our radar at all. For example, if we examine 'dangerous' intersection images and find that many of them share certain features, this provides us with new avenues to investigate in a more interpretable manner.

If our model can successfully identify intersection features related to safety, we can apply it to predict the safety of planned future intersections. Not only does this assist with intersection planning in general, but we can also use it to investigate whether safe intersections are being built equitably. We can apply our model to planned intersections in high income areas and low income areas and see whether there are any differences in safety levels—and if so, why.

With these motivations in mind, we refined three questions for our research project:

1. Can a regression model use data about intersections (speed limit, number of roads, etc.) predict how many accidents will occur at a given intersection (number of expected accidents)?
2. Does intersection street view image data improve prediction recall and precision?
3. Do higher-income areas tend to build safer intersections than their lower-income counterparts?

Methods

All analyses were performed using python in google colab jupyter .ipynb notebooks, except our image net, which was trained offline in an anaconda-based python environment. We used pytorch to build our neural nets.

Data Preparation

Our data includes information collected from the Fatality Analysis Reporting System (FARS), city-level accident datasets (OpenDataPhilly, City of Detroit Open Data Portal, NYC Open Data, and Los Angeles Open Data), OpenStreetMaps (OSM), and images collected from the Google static street view API. The primary dependent variable of our experiment is the number of accidents that occur at each intersection over a given period of time. We collected this information from 2015-2018 motor vehicle collision datasets found on OpenDataPhilly, City of Detroit Open Data Portal, NYC Open Data, and Los Angeles Open Data. In addition to this, information on traffic-collision-based-mortalities was provided by using DuckDB to scrape the 2015-2018 FARS datasets. The FARS datasets are a series of datasets containing time and location-based records of fatal car crashes in the U.S. Altogether, these datasets were downloaded from their respective databases and combined to form four complete, city-level datasets.

In order to match the collision dataset accidents to city intersections, we compared the locations of car crashes with the locations of intersections. The longitude and latitude coordinates of these intersections, along with various features of the intersection, were collected from OSM. The additional features serve as supplementary independent variables. They include: the numerical speed limits of incoming roads, incoming road types (a categorical string), road angle (a numerical value), the number of roads that feed into an intersection, and whether an intersection has a crosswalk, camera, speed display, traffic light, stop-sign, lighting, or bike lane (all individual booleans). We also used the location data to create headings and camera coordinates for our collected Google static street view images.

All of this data was collected from OSM using OSMnx, NetworkX, and GeoPandas. First, intersection node graphs from target cities (Los Angeles, New York City, Detroit, and Philadelphia) were collected using OSMnx. In these graphs, nodes represent the intersection, and the edges between nodes represent street data. Our supplementary intersection information was collected by examining the map features attached to these nodes and edges. GeoDataFrames and NetworkX were used to combine overlapping nodes and attach longitude and latitude coordinates to the intersections. This dataframe was then combined with the FARS datasets to attach the number of fatal car crashes to each intersection. This was done using a time-zone-based coordinate reference system to map the longitude and latitude coordinates of the collision datasets to the OSM data.

Our final data feature—static street view images—was collected using Google's Static Street View API. By calling this API while referencing each image's longitude, latitude, and heading (collected from OSM), we would theoretically be able to extract an image of each intersection from each of its connecting streets. However, because Google charges users 7$ for every 1000 images scraped past the first 10,000, we decided to only scrape 40,000 New York images using 4 separate google accounts, bringing us very close to the 41,773 intersections in our New York data. For this rough draft, we were only able to scrape and train our model on 10,000 New York images.

Figure 1: Examples of streetview images pulled from google (img_000001 and img_000060).

Metadata Analysis

Our final metadata database had one row for each road heading into an intersection, with the following columns:

Variable	**Description**
Intersection_id	Unique id for each intersection, each of which had many contributing roads.
Int_lat	Intersection latitude (identical for the same intersection)
Int_lon	Intersection longitude (identical for the same intersection)
Cam_lat	The latitude of the camera on the approaching road
Cam_lon	The longitude of the camera on the approaching road
Cam_heading	Camera heading, relative to north (ex: north is 0°, east is 90°, south is 180°, and west is 270°)
Speed_limit	Speed limit of the approaching road
Road_type	What type of road the approaching road is

Is_major_road	Whether or not the road was considered a ‘major’ road. These roads (which include motorways, primary roads, secondary roads, and trunks of other primary roads) are the major highways and motorways that connect cities and towns.
Has_crosswalk	Boolean whether there was a crosswalk.
Has_camera	Boolean whether the intersection has a traffic camera
Has_speed_display	Boolean whether the intersection has a speed display
Has_traffic_light	Boolean whether the intersection has a traffic light
Has_stop_sign	Boolean whether the intersection has a stop sign
Number_of_ connecting_roads	Number of roads feeding into the intersection
Is_lit	Boolean whether the road is lit. We counted NaNs as false.
Has_bike_lane	Boolean whether the road has a bike lane. We counted NaNs as false.
Num_lanes	Number of lanes of the approaching road.
Number_of_accidents	Number of accidents at the intersection (same for each intersection). This was the variable we hoped to predict.
Has_had_fatal	Boolean whether or not the intersection had seen a fatality. This was another variable we hoped to predict.

Because we were only interested in intersections, we collapsed the data from incoming roads into a single road per intersection. Most columns at the same intersection were identical, or were true for the intersection if they were true for any connecting road. For road-specific columns, we took the maximum value for speed_limit, is_major_road, number_of_connecting_roads, num_lanes, and the first for camera variables, road_type, and has_bike_lane.

Because much of our metadata was crowdsourced, we were concerned that some variables such as the presence of a traffic light might be under-reported. We compared the number of traffic lights present in our NYC dataset (12289) to the actual number of traffic lights in NYC (13543 as of March 2022) (NYC DOT - Infrastructure - Traffic Signals, n.d.). This shows that while some traffic lights are missing, the majority of intersections have probably been reported correctly. With this confirmed, we felt comfortable proceeding with our analysis.

While our ultimate goal was to build an image net, we recognized that this would be very uninterpretable, and it would be difficult to compare its efficacy to other models. For those reasons, we began by assembling a database of metadata about intersections, and training a number of models on that information. This way we could explore how accurate a metadata model could be without using images, and how much accuracy increases with the addition of images. We can also compare our black box image model to the interpretable metadata model to see if we can guess what the image model might be doing. Building a metadata model first also let us build a simpler model before moving on to the CNN.

We made a few more adjustments to our metadata database to prepare it for analysis:

- Because we were interested in accidents at intersections instead of contributing roads, we combined rows representing the same intersection into a single row per intersection.
- We removed data we did not intend to train our model on, such as latitude and longitude, and has_had_fatal.
- The final list of columns we used to train our metadata model was: speed_limit, road_type, is_major_road, has_crosswalk, has_camera, has_speed_display, has_traffic_light, has_stop_sigh, number_of_connecting_roads, num_lanes.

Small Models

We ran a number of small models on our metadata model to explore which algorithms would be most effective. Using number_of_accidents as the outcome variable, we built and trained several linear models on our metadata dataset, to compare them and see which ones performed best. We used the sklearn score() method to calculate R^2.

After seeing these results, we ran a number of exploratory follow up tests to improve our results, increase generalizability, and increase interpretability. These investigations are detailed in our results section.

Neural network

Using pytorch, we built a simple neural network with four fully connected layers, each run through a ReLU layer before being passed on to the next. We made a custom dataset class for our metadata database. Then we split the data, trained the model for 5 epochs, printing the loss after every 200 batches.

Metadata and Image Analysis

We modified our dataframe to include two additional columns: ‘path’, with a path to the needed image, and index, used to match images of incoming roads to the appropriate row in the collapsed intersection metadata dataframe. To reduce training time and model complexity, we only used an image of one incoming road for each intersection.

We picked 10,000 random New York intersections, then downloaded images of these intersections from Google static street view API to github. Once downloaded to github, we could access these images indefinitely without paying.

We made a custom pytorch Dataset for our image dataset. Then we built a convolutional neural network, which takes in 400 x 400 x 3 images to predict number_of_accidents at an intersection.

For our final draft, we plan to feed the output of this network for each intersection image as another input to our metadata model.

Results

Metadata Analysis

We ran several simple linear models on our metadata, with the following result:

Model	**Score (R^2)**
Linear Regression Model	0.2967
Lasso Model	0.2699
Ridge Model	0.2967
Elastic Model	0.2167
Histogram-based Gradient Boost Model	0.3781
SGD Model	0.2943

Most of the models were similar, however the gradient boost model performed better than any of the others. We calculated the RMSE of this model (for comparison with neural network) and found it was 14.38. With an average of 8.3 accidents per intersection, a mean error of almost 15 accidents is not good.

City-Specific Model

The accuracy of these models was very low. We considered that some cities might have different consistent patterns in what kinds of intersections are dangerous. To check this, we built and tested models on New York City alone to see if they were more effective. We only used the linear model and gradient boost model, as the other models did not show significant differences from each other.

Model	**Score (R^2)**
Linear Regression Model	0.3183
Histogram-based Gradient Boost Model	0.3810

The New York City model performed better than the joint model, but still not very well. This suggests that there are indeed consistent differences in intersection safety practices between cities, but that these differences are not sufficient to account for the failure of our multi-city models.

City Generalizability

To further explore the differences between cities and generalizability of our model, we trained the linear model and the gradient boost model (the most effective one) on New York City, Detroit, and Philadelphia, and tested it on Los Angeles.

Model	**Score (R^2)**
Linear Regression Model	0.2294
Histogram-based Gradient Boost Model	-0.1917

We were very surprised to see that the gradient boost model was actually worse than guessing in this situation. Possibly Los Angeles is sufficiently different from the other three cities that training a model on them makes it completely inadequate for guessing intersection danger in Los Angeles.

Location Information

We considered that the location of the intersection might actually play an important role in whether or not the intersection was dangerous. Possibly some parts of cities have worse wear from weather conditions, or some neighborhoods have less safe driving conditions. We added longitude and latitude back to the data, and reran our models for New York City.

Model:	**Score (R^2)**

Linear Regression Model	0.4788
Histogram-based Gradient Boost Model	0.3357

The linear model was much more accurate with locations, though the gradient boost model was slightly worse. We wanted to see what variables the model was paying most attention to, and how these differed for the two models. So we printed out the weights for each column, for the linear models with and without location data.

Column	**Coefficient (no locations)**	**Coefficient (with locations)**
int_lat	0.0	NA
int_lon	5.39	NA
cam_lat	0.0	NA
cam_lon	-2.37	NA
cam_heading	0.0	NA
speed_limit	0.23	0.16
road_type	0.67	1.12
is_major_road	0.01	0.0
has_crosswalk	-4.13	1.8
has_camera	0.0	0.0
has_speed_display	0.0	0.0
has_traffic_light	11.99	14.22
has_stop_sign	1.33	2.11
number_of_connecting_roads	7.75	4.93
num_lanes	3.22	1.46

This suggests that location only matters somewhat. The coefficients for latitude were 0. The coefficients for longitude were larger, but in opposite directions: int_lon had a positive coefficient of 5.39 while cam_long had a negative coefficient of -2.37. These results would cancel each other out to a large degree. Additionally, it seems strange that longitude would matter but not latitude. Possibly this could be due to the unique geography of New York City.

Another notable feature of the table is the high impact of traffic light presence: this term has a coefficient of 12 and 14, with the second highest term being only 7. We explore this further in our discussion.

Worth noting is also the significant differences between the coefficients of some variables from one model to the next. For instance, with location data crosswalk presence had a coefficient of -4.13, but without location data this rose to 1.8. The influence of number_of_connecting_roads also changed significantly. It seems unlikely that these features have genuine interactions with longitude: more likely neither of these models is very robust.

None of the linear models we built with our metadata performed particularly well. We hoped our planned metadata neural network might be able to find some nonlinear relationships in the data and make more accurate predictions.

Metadata Neural Network

Our model's loss went down from about 0.7 to 0.55 in the first epoch, but did not improve at all over the course of the next four epochs, so we did not feel the need to run it for longer than five epochs. We evaluated our neural network against the test dataset. Our RMSE was 2.5 (again, with a mean of 8 accidents), which is significantly more accurate than the gradient boost model.

We used LIME to investigate the effects of each feature on final predictions. We found that whether or not an intersection had a traffic light was by far the most impactful. Intersections with traffic lights were regularly predicted to have 14 or more accidents, while those without were predicted to have 5 or less, with only a few exceptions. We discuss the implications of this in our discussion.

Image and Metadata Neural Network

We will add this section for our final draft once we have a chance to test our image model and analyze the output.

Discussion

None of the models we created showed much ability to predict intersection danger, and the few that did appeared to be drawing conclusions from features that merely correlate with greater numbers of accidents (such as traffic light presence) that would not be useful for urban planning.

However, this does not mean we learned nothing from our investigation. For instance, several of our investigations suggest that there are differences in intersection safety from one city to the next. Models trained on New York data alone were better at predicting accident totals in New York, despite having trained on less data. Meanwhile, generalizing to Los Angeles from the other three cities made significantly worse predictions. All this suggests that consistent differences between cities may play an important role in intersection safety, and we should be cautious about generalizing any conclusions about traffic safety from one city to the next. This is particularly important to keep in mind in the context of the literature from our introduction. As models continue to be trained and published using data from a wide variety of locations: it may be prudent to avoid using models trained in one region to make predictions in another region until further research can be done on how applicable models trained on one city's data are to another.

Influence of Traffic Lights

One result we did not expect was the strong influence traffic light presence had on the number of predicted accidents. Intersections with traffic lights were consistently predicted to be far more dangerous than those without.

We did an exploratory investigation into why this might be the case. Inferring a causal relationship between traffic lights and accidents seems intuitively wrong: traffic lights are added to intersections to prevent accidents. Yet, there is a correlation between traffic light presence and intersection danger in our data. We spent some time looking into why this was the case.

We considered that the effect might be due to the aforementioned consistent difference between cities. New York City has significantly more traffic lights than any other city (12289 compared to 3413 in LA). We also found that in New York City there were significantly more accidents (15 per intersection, where LA had only 5), and that at intersections with traffic lights, New York had a mean of 32 accidents and intersections without traffic lights had a mean of only 8.

What we took away from this was that traffic lights were more prevalent in New York, and that more accidents occurred in New York in general. This might bias our model to use traffic lights as a proxy for the more dangerous New York City intersections. To follow up on this, we trained our linear model on non-NYC data to see if it treated the traffic lights differently. We found that has_traffic_light still had the strongest impact with a model weight of 9.67. This suggests that the impact of traffic lights was not an artifact of traffic lights being more prevalent in New York city.

Our next theory was that because traffic lights are added to busier intersections, traffic light presence is acting as a proxy for the amount of traffic flow through an intersection. We would expect intersections with higher traffic flow to see more accidents. To test this theory, we adjusted our outputs using AADT (annual average daily traffic). We divided the number of accidents by the AADT (or by 500 if it was below 500, to avoid outliers throwing off the model), then multiplied by 1000 to bring the adjusted value to a more interpretable number. Then we trained the linear model on the adjusted data. In this model we again found that has_traffic_light was still significantly more impactful than any other input, with its model weight of 3.4 much higher than the second highest model weight of 1.89 for number_of_connecting_roads. Its R^2 score was also only 0.1258, which was worse than the other linear models.

We considered that even if using AADT to normalize the number of accidents really is a better way to assess intersection danger, our linear model might not be able to capture the complex relationships of this data as well. We trained a neural network on the normalized data. We trained for 20 epochs, but the loss did not decrease. When we evaluated our model, our RMSE was 1.5420. Given that the mean number of accidents after adjustment was roughly 4.28, this is not a great error. Using the LIME explainer on this net, we found that has_traffic_light was still one of the most impactful variables in most decisions (with speed_limit also having high impact).

We assumed going in that intersections with a higher traffic flow would have more accidents. This makes intuitive sense: if there are more cars in an intersection there is more of a chance for accidents to occur at all, and the higher car volume likely means a single car going through the intersection is more likely to get into an accident while there. To confirm our intuition, we decided to plot traffic flow against number_of_accidents and run a linear regression. When we did this, we were surprised to find that traffic flow and number of accidents do not appear to correlate at all (figure 2). In fact, the best fit line we plotted actually decreases slightly.

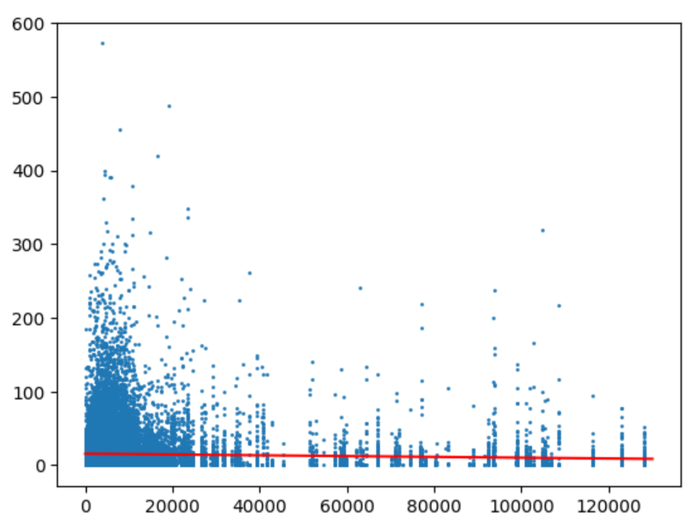

Figure 2: Traffic flow (AADT) plotted against number_of_accidents.

Possibly something is wrong with our data and mapping of AADT to intersections. It's also possible that some high volume intersections are part of motorways that have more safeguards. If this is not a coding issue, our dataset does not contain enough information to investigate why this is the case. We would likely have to look deeper into specific intersections, including some data not captured by any of our datasets, to see why this might be the case.

In any case, it is presently unclear to us why traffic light presence is such a determinant for intersection danger. This is one major area for future work on this project to follow up on.

Limitations & Future Directions

Our project has many limitations:

City-Specificity: We only took intersections from New York, Los Angeles, Detroit, and Philadelphia. Given the evidence suggesting city matters a great deal in predicting intersection danger, at best our model could only be generalized without extreme caution to these four cities. At worst it may not be effective on any of these cities, if the ways in which they vary prevent the model from learning consistent patterns.

Flawed Data: While we did take steps to ensure the crowdsourced data we used was complete enough for our purposes, we still noticed a number of strange things about the final dataset, including some rows whose number of lanes was marked as 0. Possibly some strange features of our dataset may have hindered our model, and we would need to engage in more thorough data cleaning to make the best possible model.

Incomplete Feature List: As discussed in our introduction and other literature on the subject, there are many variables feeding into intersection data that our metadata model did not capture, such as presence gas stations, and road width. We hoped that our image-based CNN might be able to combat some of these limitations, but we have had no luck with this so far.

More broadly, our investigations are limited in that they lack predictive power. As seen with our exploration of the impact of traffic lights, our models are highly sensitive to features that merely correlate with accident totals rather than contribute to them.

One major takeaway from these limitations is reinforcing the importance to machine learning of having a complete, relevant, and high-quality dataset. While our metadata dataset was very large and exhaustively covered a number of features we believed might contribute to intersection safety, it appears that it did not include enough information to make useful predictions based on intersection features. Again, we hoped that including image data would account for the limits of our metadata, but this did not prove to be the case. The most fruitful future development of this project would likely involve acquiring more metadata that we have reason to believe might be relevant to intersection safety (such as road width and commercial features (Yang et al. 2024)); or potentially creating a more precise image CNN to identify and target specific intersection features we want to investigate.

Social & Ethical Implications

We originally hoped to test our model on upcoming intersection designs in low income and high income areas, to see where the safest and most dangerous intersections are being built. Given our model's failure to predict intersection danger using design features, we did not believe this would tell us anything useful, and did not pursue it.

We have no reason to believe our model is at risk of perpetuating any injustices. Our dataset did not include any information on sensitive factors. However, if effective intersection danger prediction models were to be developed in the future, it would be worth using a tool such as the fairlearn package to investigate the model using factors such as neighborhood income level of a particular intersection as a sensitive_feature. This way we could determine with more confidence whether or not our model perpetuates any systemic issues.

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